Chapter 5 IT Infrastructure and Emerging Technologies

Learning Objectives

After studying this chapter, students should be able to answer the following questions:

- What is IT infrastructure, and what are the stages and drivers of IT infrastructure evolution?
- What are the components of IT infrastructure?
- What are the current trends in computer hardware platforms?
- What are the current computer software platforms and trends?
- What are the challenges of managing IT infrastructure, and what are the management solutions?

5.1 What Is IT Infrastructure, and What Are the Stages and Drivers of IT Infrastructure Evolution?

IT infrastructure means the shared technology resources that provide the platform for a firm's information system applications. It includes hardware, software, and services (such as consulting, education, and training) that are shared across the whole firm or across whole business units. You can think of it as the foundation on which everything else in the firm's IT is built. It supports how the firm serves customers, works with suppliers, and runs its internal processes.

Infrastructure is not only machines. It also includes a set of services, made of both people and technology, that management pays for. These are listed below.

Services that make up IT infrastructure

Infrastructure service	What it provides
Computing platforms	Mainframes, midrange computers, desktops, laptops, mobile devices, and cloud computing that connect employees, customers, and suppliers.
Telecommunications services	Data, voice, and video connections for employees, customers, and suppliers.
Data management services	Storing and managing company data, and tools for analyzing it.

Infrastructure service	What it provides
Application software services	Firmwide software such as ERP, CRM, supply chain, and knowledge management systems.
Physical facilities management	Building and managing the physical sites needed for computing, telecommunications, and data management.
IT management services	Planning the infrastructure, working with business units, accounting for IT spending, and project management.
IT standards services	Policies on which technology will be used, when, and how.
IT education services	Training employees in using systems, and training managers in planning and managing IT investments.
IT research and development	Research on future IT projects and investments that could help the firm stand out in the market.

Technology Drivers of Infrastructure Evolution

Infrastructure has changed because of improvements in processors, memory chips, storage, networking, and software design, which made computing power grow very fast while cost fell very fast. The main drivers are below.

- ***Moore's Law and Microprocessing Power***

In 1965, Gordon Moore observed that the number of transistors on a microprocessor chip doubles approximately every year. Later, he revised this prediction to every two years. A transistor is a tiny electronic switch inside a processor. The more transistors manufacturers can fit onto a chip, the more processing capability they can achieve. As a result, computers become more powerful, smaller, and relatively cheaper over time. Three common variations of Moore's Law:

- Microprocessor power doubles every 18 months.
- Computing power doubles every 18 months.
- The cost of computing falls by half every 18 months.

Nanotechnology is the technology of working with individual atoms and molecules to create extremely small devices and components.

Is Moore’s Law Still Relevant Today? Yes. Moore’s Law is still relevant, but its traditional growth rate has slowed down. Manufacturers are finding it increasingly difficult and expensive to make transistors smaller. However, they continue to improve processor performance through advanced chip designs and new technologies. For example: Today's processors contain billions of transistors and support advanced applications such as Artificial Intelligence (AI).

- ***The Law of Mass Digital Storage***

The amount of digital information is roughly doubling every year, while the cost of storing it is continuously decreasing.

- Digital data is growing rapidly due to social media, online transactions, videos, and business activities.
- Storage capacity for the same amount of money is increasing over time.
- This allows individuals and businesses to store more data at a lower cost.

- ***Metcalfe's Law and Network Economics***

Metcalfe’s Law states that the value of a network increases as more people join and connect to it.

- More users joining a network create more opportunities for communication and connection.
- As the number of users increases, the value of the entire network grows rapidly.
- This encourages people and businesses to use more computing resources and network technologies.

Example: Think about Facebook. If only 10 people use Facebook, its value is limited. But when millions of people join, users can connect, communicate, share information, and do business with each other. This makes the network more valuable.

- ***Declining Communication Costs and the Internet***

The cost of communication is decreasing rapidly, while the number of Internet users is growing exponentially.

  - Internet and telephone communication are becoming cheaper.
  - As communication costs approach zero, the use of communication and computing facilities increases rapidly.
  - Businesses are expanding their Internet connections, wireless networks, client/server networks, desktop computers, and mobile devices to benefit from the Internet.

- ***Technology Standards and Network Effects***

Technology standards are specifications that make sure products work together and can communicate over a network. Standards create economies of scale: manufacturers focus on products built to one standard, so prices fall. Without them, computing would be far more expensive. Some important standards are shown below.

Table 5.1 Some important standards in computing

Standard	Why it matters
ASCII (1958)	Let computers from different makers exchange data. Later became the universal language linking keyboards and mice to computers.
Unix (1969 to 1975)	A powerful, portable, multiuser operating system that runs on many makers' computers. Became the most widely used enterprise operating system.
TCP/IP (1974)	The set of rules and addressing scheme that connects millions of computers into the Internet.
World Wide Web (1989 to 1993)	Standards for storing, finding, and displaying information as linked pages with text, graphics, audio, and video.

5.2 What Are the Components of IT Infrastructure?

IT infrastructure today has **seven major components**. They are investments that must be coordinated so that the firm gets one coherent infrastructure.

The seven components of IT infrastructure

Component	What it covers	Examples of vendors
1. Computer hardware platforms	Mainframes, servers, PCs, tablets, and smartphones.	Intel, AMD (chips); IBM, Dell, HP, Apple
2. Operating system platforms	Software that manages the computer's resources. For servers: Windows Server, Unix, Linux. For clients: Windows, Mac OS, Chrome OS, Android, iOS.	Microsoft, IBM, HP, Oracle, Google, Apple
3. Enterprise software applications	ERP, CRM, supply chain, and other firmwide applications, plus middleware that links existing systems.	SAP, Oracle, Microsoft, IBM
4. Data management and storage	Database software to organize and manage data, and physical storage for large systems.	IBM (DB2), Oracle, Microsoft SQL Server, MySQL, Hadoop; EMC (storage)
5. Networking and telecommunications	Network operating systems, networking hardware, and telecom services for voice, data, wireless, and Internet.	Cisco, Juniper; AT&T, Verizon
6. Internet platforms	Hardware, software, and services to support a firm's website, including web hosting, routers, and cabling. Also web development tools.	Microsoft (.NET), Oracle-Sun (Java), Adobe, IBM, Dell, HP

Component	What it covers	Examples of vendors
7. Consulting and system integration services	Experts who help plan, install, and train staff for new infrastructure.	Accenture, IBM, HP, Infosys, Wipro

5.3 What Are the Current Trends in Computer Hardware Platforms?

Computing power is moving onto networks and mobile devices. The book looks at seven hardware trends: the mobile digital platform, consumerization of IT and BYOD, quantum computing, virtualization, cloud computing, green computing, and high-performance and power-saving processors.

1. The Mobile Digital Platform

Mobile digital platforms, such as smartphones and tablets, have emerged as alternatives to traditional PCs and larger computers.

- Smartphones (iPhone and Android) can perform many PC functions, such as browsing the Internet, sending emails, messaging, and accessing corporate data.
- Other mobile devices include netbooks, tablets (iPad), and e-book readers (Kindle).
- Businesses use smartphones and tablets to access sales information, financial reports, manufacturing data, and project updates.
- Wearable devices, such as smartwatches, smart glasses, smart ID badges, and fitness trackers, are also becoming part of mobile computing.

2. Consumerization of IT and BYOD

BYOD (Bring Your Own Device) is the practice where employees use their personal devices, such as smartphones, tablets, and laptops, for work purposes. Consumerization of IT is the process through which technologies and services first developed for individual consumers become widely used in businesses. Examples: Gmail, Google Apps, Dropbox, Facebook, and Twitter.

Impact on Businesses:

- Employees increasingly want to use their personal devices to access company systems.
- Businesses have less control over the selection and use of technology.
- IT departments face challenges in maintaining security, protecting company data, and managing different devices.
- Technology decisions are no longer controlled only by the IT department; employees and business departments also influence them.

3. Quantum Computing

A normal computer handles a bit of data as either 0 or 1. Quantum computing uses the principles of quantum physics so that a bit can be 0, 1, or both at the same time. This gives it huge processing power and could solve some problems millions of times faster than today's computers, in cases where normal computers would need many years. IBM has made quantum computing available to the public through IBM Cloud, and that Google's parent company Alphabet and Lockheed Martin use quantum platforms. It is still an emerging technology.

4. Virtualization

Virtualization means presenting computing resources, such as processing power or storage, so they can be used without being limited by physical setup or location. One physical machine can look like many logical machines. For example, one server can run many copies of an operating system, and each "virtual server" looks like a real server to the software. VMware is the leading vendor for Windows and Linux servers.

- **Saves money.** Most servers run at only 15 to 20 percent of capacity. Virtualization can raise this to 70 percent or more, so fewer computers do the same work, with less space and less energy.
- **Easier administration.** Hardware can be managed from one place.
- **The opposite direction also works.** Several physical resources can appear as one. In **software-defined storage (SDS)**, software for managing storage is separated from the storage hardware. Firms can pool storage, use cheaper standard hardware or cloud storage, and avoid wasting capacity.

5. Cloud Computing

Cloud computing means a company or person does their computing on a virtualized infrastructure in a remote place, and uses it through a network, mainly the Internet.

Essential characteristics of cloud computing

Characteristic	Meaning
On-demand self-service	Users can get computing capability, such as server time or storage, automatically as needed, without help from the provider.
Ubiquitous network access	Resources can be reached with standard network and Internet devices, including mobile devices.
Location-independent resource pooling	Resources are pooled to serve many users and assigned dynamically as needed. The user generally does not know where they are located.
Rapid elasticity	Resources can be quickly increased or reduced to match demand.
Measured service	Charges are based on the amount actually used.

Three Types of Cloud Services

Service	What the customer gets	Examples
Infrastructure as a service (IaaS)	Processing, storage, networking, and other computing resources, to run their own information systems. Pay only for what is used.	Amazon Web Services: S3 for storage and EC2 for running applications

Service	What the customer gets	Examples
Software as a service (SaaS)	Software hosted by the vendor and delivered over a network. Users open it in a web browser. Usually paid by subscription.	Google Apps, Salesforce.com
Platform as a service (PaaS)	Infrastructure and programming tools from the provider, so customers can build and run their own applications.	IBM Bluemix, Salesforce Force.com

Public, Private, and Hybrid Clouds

Table 5.2 Cloud computing models compared

Type	Description	Managed by	Best for
Public cloud	Owned by a service provider and offered to the general public or an industry group. Examples: AWS, Google Drive, Dropbox, iCloud.	Third-party providers	Firms without major privacy concerns, pay-as-you-go needs, or limited IT resources and skills.
Private cloud	Operated only for one organization. Can be hosted inside or outside the firm.	In-house IT or a private third-party host	Firms with strict privacy and security needs, or that must control where their data is kept.
Hybrid cloud	A combination of private and public cloud services that remain separate.	In-house IT, private host, and third-party providers	Firms that want some in-house control but also use public cloud for part of their IT.

Because public cloud users do not own the infrastructure, they do not need large investments in hardware and software. They pay for what they use (**utility computing**) or by monthly or yearly subscription. This is also called **on-demand computing**. Large firms usually prefer a **hybrid model**: their own infrastructure for the most essential core activities, and public cloud for less critical systems or extra capacity in busy periods. Startups and small firms with small budgets gain the most from public cloud.

Drawbacks. Unless data is also stored locally, the provider controls the data. Some companies worry about the security of putting critical data in the hands of an outside vendor that also serves other companies. Companies also expect systems to run 24/7 and do not want to lose business if the cloud fails. Even so, the trend is clearly toward more cloud use.

6. Green Computing

Green computing (or green IT) means practices and technologies for designing, making, using, and disposing of computers and related equipment so that the damage to the environment is as small as possible. The world's data centers use as much energy as 30 nuclear power plants, about 1.5 percent of all energy used in the world, and that one corporate data center can use over 100 times the power of a normal office building. Virtualization is one of the main ways to cut this, because it reduces the number of machines and the power they use. New data centers are designed for energy efficiency, and companies such as Microsoft, Google, Facebook, and Apple are using clean energy such as wind and hydropower.

5.4 What Are the Current Computer Software Platforms and Trends?

Modern software platforms focus on open-source technologies, cross-platform applications, system integration, outsourcing, cloud services, and mobile applications. There are four major trends in software platform development:

1. Linux and Open-Source Software

Open-source software’s source code is freely available, allowing users to use, modify, and distribute it. Linux is a popular open-source operating system used in servers, smartphones, and supercomputers. Examples: Linux, Mozilla Firefox, and Apache.

2. Java, HTML, and HTML5

- Java: A programming language that allows applications to run on different operating systems and devices.
- HTML: A markup language used to create and structure webpages containing text, images, links, and other content.
- HTML5: An advanced version of HTML that supports audio, video, and other multimedia content directly on webpages without additional plugins.

3. Web Services and Service-Oriented Architecture (SOA)

- Web Services: Software components that allow different applications to communicate and exchange information, even if they use different operating systems or programming languages.
- XML (Extensible Markup Language): A language used to store, organize, and exchange data between different systems.
- SOA (Service-Oriented Architecture): A software design approach where applications are built by combining different independent services that communicate with each other.

Example: A customer books a flight through an airline website and also reserves a rental car from another company through the same website. Web services allow both companies' systems to exchange information.

4. Software Outsourcing and Cloud Services

Businesses can obtain software from external providers instead of developing everything themselves.

Three sources of external software:

- Software Packages: Ready-made software purchased from vendors to perform specific business functions, such as payroll and accounting.
- Software Outsourcing: Hiring an external company to develop, maintain, or manage software.
- Cloud-Based Software Services: Using software over the Internet without installing and maintaining it on individual computers.

SaaS (Software as a Service): Software delivered over the Internet, usually through a subscription or pay-per-use model. Example: Salesforce, Google Workspace.

SLA (Service Level Agreement): A formal agreement between a service provider and customer that defines the services, responsibilities, performance standards, and support expected.

5. Mashups and Apps

Mashups: Applications created by combining features or data from two or more different applications to provide additional value. Example: A real estate website combining Google Maps with property listings to show property locations.

Apps: Small, specialized software programs designed mainly for mobile devices such as smartphones and tablets. Examples: Facebook, Google Maps, WhatsApp, and banking apps.

5.5 What Are the Challenges of Managing IT Infrastructure, and What Are the Management Solutions?

Managing a coherent infrastructure raises three kinds of challenge: dealing with platform and infrastructure change, management and governance, and making wise infrastructure investments.

- **Dealing with Platform and Infrastructure Change**

Scalability refers to the ability of an IT system to handle increasing users and workloads without breaking down. Businesses must plan their IT capacity to support growth, new applications, and changing workloads. They also need proper policies to manage and secure mobile devices and cloud services. Management must ensure that critical systems are reliable, secure, available, and provide acceptable response times.

- **Management and Governance**

IT Governance determines who controls and manages a firm's IT infrastructure. It addresses whether IT decisions should be made centrally or by individual departments, how IT responsibilities are shared between central management and business units, and how infrastructure costs are distributed. Each organization makes these decisions based on its needs.?

- **Making Wise Infrastructure Investments**

Infrastructure is a major investment, and it is hard to get the amount right. If a firm spends too much, the infrastructure sits idle and hurts financial performance. If it spends too little, important services cannot be delivered and competitors who spent the right amount will do better.

A related question is the **rent-versus-buy decision**: should the firm buy and maintain its own infrastructure or rent from outside suppliers, including cloud providers? Cloud computing is a low-cost way to gain scalability and flexibility, and it can reduce hardware, software, staff, and maintenance costs and let a firm focus on its core business. But the firm should check security needs and the effect on business processes. In some cases, renting software costs more in the long run than buying and maintaining it in-house. The book notes that cloud reliability has improved a lot, and in 2016 some Fortune 500 firms planned to move most of their computing to the cloud.

Total Cost of Ownership (TCO)

The real cost of technology is more than the purchase price. The **total cost of ownership (TCO)** model looks at all direct and indirect costs, including buying and installing, and the continuing costs of administration, upgrades, maintenance, support, training, and even power and space.

Table 5.4 Total cost of ownership (TCO) cost components

Cost component	What it includes
Hardware acquisition	Purchase price of computers, terminals, storage, and printers.
Software acquisition	Purchase or license of software for each user.
Installation	Cost of installing computers and software.
Training	Training for IS specialists and end users.
Support	Ongoing technical support and help desks.
Maintenance	Upgrading hardware and software.
Infrastructure	Networks and special equipment such as backup storage units.

Cost component	What it includes
Downtime	Lost productivity when hardware or software failures stop users from working.
Space and energy	Real estate and utility costs of housing and powering the technology.

To cut TCO, firms can move to cloud services and use more centralization and standardization: fewer types of computers and software, administered and fixed from one central location, which also reduces the support staff needed. For mobile devices, the productivity gains must be balanced against the costs of integrating them and supporting them, including wireless fees, training, and help desk support.

Review Questions

1. Define IT infrastructure from both a technology and a services perspective.
2. List each era in IT infrastructure evolution and describe its main features.
3. Define and describe a web server, an application server, and a multitiered client/server architecture.
4. Describe Moore's Law and the Law of Mass Digital Storage.
5. How do network economics, declining communication costs, and technology standards affect IT infrastructure?
6. List and describe the components of IT infrastructure that firms need to manage.
7. Describe the mobile digital platform, consumerization of IT, and cloud computing.
8. Explain how businesses can benefit from virtualization, green computing, and multicore processors.
9. List the essential characteristics of cloud computing and describe the three types of cloud services.

10. Define open source software and Linux, and explain their business benefits.
11. Define Java and HTML5 and explain why they are important.
12. Define web services and explain the role of XML.
13. Name and describe the three external sources of software.
14. Define and describe software mashups and apps.
15. Name and describe the management challenges posed by IT infrastructure.
16. Explain how the competitive forces model and the TCO of technology assets help firms make good infrastructure investments.